

LinkForm: a Multipurpose VR-based Remote Skill-Transfer Robotic System

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ABSTRACT

Physical interaction plays a vital role in various skill-transfer scenarios. Despite its importance, the majority of systems focus on visual and auditory in immersive environments for training contexts. Therefore, we present LinkForm, a novel VR-based robotic skill-transfer system comprises a VR environment and a robotic system. LinkForm enables a coach to remotely train a user, with the ability to convey visual, verbal, and physical guidance and interactions to the remote trainee. The preliminary evaluation result shows statistically significant improvements in the participants performance after using LinkForm. Participants also showed strong interest in training using LinkForm, suggesting various remote-training scenarios.

Index Terms: VR, Robot, Coaching System, Metaverse.

1 INTRODUCTION

Remote collaboration systems have proved to play a vital role during the global coronavirus pandemic. Despite the wide availability of various remote collaboration and teleconferencing systems, most systems solely focus on visual and auditory communication mediums, which are limited. For example, Hawkar Oagaz et al. [7] proposed methods to enhance table tennis skills using Virtual Reality (VR), emphasizing visual and auditory cues. Despite their effectiveness in skill-transfer applications, most VR-based skill-transfer systems fall short in enabling physical interactions and guidance to trainees, which are pivotal for effective skill-transfer [3, 5, 7, 8].

To address this gap, we propose *LinkForm*, a multipurpose remote skill-transfer system consisting of a robot arm and a VR system. *LinkForm* consists of two subsystems: *trainee system*: snake-shaped robot and a VR system *Coach System*: a VR system to control both the robot and interact with the trainee. LinkForm advances the state of the art by being the first multipurpose robotic system designed explicitly for skill-transfer applications. We demonstrate various potential applications of LinkForm for remote skill transfer, including drawing and kendo. We conducted a user study involving 12 participants investigating the skill-transfer effect. The results show that the LinkForm enabled subjects to improve their level of basic table tennis skills, highlighting the need for further exploration into the potential of this system.

We summarize our contributions as follows: 1) We Present the design and implementation of a novel skill-transfer system, *LinkForm*, and demonstrate it's potential skill-transfer applications. 2)

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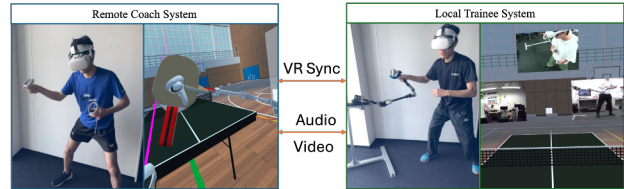


Figure 1: LinkForm is a novel skill-transfer system that uses VR and a robot to enable physical guidance of a trainee's body posture and hand motion in real time by an expert.

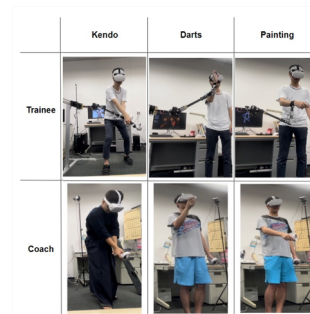


Figure 2: We developed several training scenarios in various sports, including Kendo, Darts, and Painting as they require both precise and specific hand motions and body postures, which are the main skill-transfer capabilities in LinkForm.

present preliminary evaluation results that show the effectiveness of LinkForm and potential of LinkForm.

2 DESIGN AND IMPLEMENTATION OF LINKFORM

To enable a coaching experience involving a coach and a trainee using LinkForm, our implementation includes two main VR environments for each the coach and trainee (fig. 1), and a robot system on the trainee side that is controlled by the coach. We explain these systems as follows:

2.1 Trainee and Coach VR Systems

The Trainee and Coach Systems each comprise a VR subsystem deployed on the Meta Quest 2 HMD [4]. The VR subsystem includes a virtual world with virtual training apparatus (e.g. racket, balls, sword.etc), where such apparatus's motions and locations are shared and synchronized between both the trainee and coach systems, similar to multiplayer games. The Trainee and Coach VR systems are implemented using Unity3D and networked connected in real time using Photon Unity Networking 2 (PUN2).

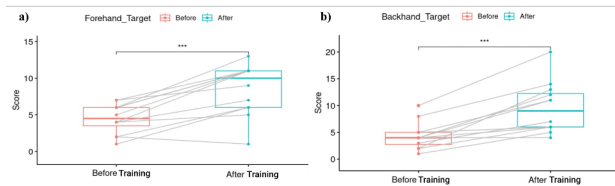


Figure 3: a) Results of Forehand.Target before and after training. b) Results of Backhand.Target before and after training.

2.2 LinkForm's Robot System

The Trainee system includes a snake-shaped robot system to guide the trainee's hand motions in different training scenarios, which is remotely controlled over the network by the Coach. The coach can use the VR joystick to engage with the VR environment and remotely control the robot motions in different scenarios. The LinkForm robot consists of six interlinked servomotors using aluminum and PLA frames, which is mounted on a metal pole to enhance stability and has 9 DoF and a changable end-effector for versatility. The robot is controlled over the network using a custom robot-control middleware [1, 2]. Therefore, LinkForm can be customized for use in a variety of skill-transfer scenarios as shown in Fig 2.

3 PRELIMINARY EVALUATION

Objective: Our study had two main objectives: 1) to evaluate LinkForm's effectiveness in a realistic skill-transfer scenario and 2) to gauge user impressions and opinions about using LinkForm. Therefore, we chose two table tennis training routines, which were chosen based on consultation with an expert table tennis trainer. We chose table tennis as it is relatively easy to learn and was thoroughly used in a variety of previous research to evaluate systems [5, 7, 8].

Participants: We recruited 12 participants (11 males, 1 female, age $m=23.6$, $SD=1.62$), who were novice table tennis players and did not report health issues that hindered their system usage.

The skill-Check tests included two tests. Both tests started with experimenter launching a ball with fixed speed, height, rotation, and direction. The first test required returning the ball to either corners of the tennis table using a forehand stroke, while the second test required using a backhand stroke. The tests were conducted 20 times for each forehand and backhand strokes; 10 trials before and 10 after training with LinkForm in each condition (total of 40 trials).

Flow: Participants were first introduced to the experiment procedure, then they took the Skill-Check-Test to assess their table tennis skills before using LinkForm. They then engaged in LinkForm training in two phases, the first consisting of training with the robot guidance in VR, and the second had verbal and visual guidance by the coach in VR only. Then, participants retook the Skill-Check-Test. Lastly, participants took a questionnaire (6-point likert scale, 6 is strong agreement) to explore their opinions and impressions.

4 RESULTS

4.1 Skill-Check Test Results

The results show statistically significant improvements in participants' performance (Fig. 3). The detailed results are as follows:

Forehand Stroke: Prior to system usage, the average score was 4.91 ($SD = 2.02$). After system usage, the average score rose to 9.27 ($SD = 3.50$). Out of the 12 participants, 11 showed score enhancements. The average change in score was 4.36 ($SD = 2.41$). The paired-sample t-test resulted in a p-value of 0.000647.

Backhand Stroke: Prior to system usage, the average score was 4.72 ($SD = 2.53$). After system usage, the average score rose to 10.5 ($SD = 4.74$). Out of the 12 participants, 11 exhibited score

enhancements. The average change in score was 5.73 ($SD = 3.41$). The paired-sample t-test resulted in a p-value of 0.000121.

4.2 Qualitative Results

Overall, participants thought LinkForm improved their table tennis skills ($m=4.75$, $SD = 0.87$) and increased their motivation for training ($m=4.92$, $SD = 1.38$). A participant mentioned "Using the robot to learn hand movements, in addition to verbal guidance, made understanding form easier" and another "The ability to learn remotely was beneficial". Although participants thought the system was safe to use ($m=1.92$, $SD = 0.793$), they voiced their concerns about several issues such as VR-motion sickness after prolonged use, and arm-movement limitations due to the robot's fixed location. Lastly, participants strongly wanted to use LinkForm for future training scenarios in other activities and sports ($m= 5.08$, $SD = 0.792$), such as painting or Kendo.

5 CONCLUSION

LinkForm presents novel potentials for skill-transfer across various contexts. The novelty of LinkForm lies in the use of shared VR between the coach and trainee, visual feedback to maintain correct body posture, and a robot arm that can physically be used to guide the trainee's hands during skill-transfer. We demonstrate the versatility of LinkForm by developing a variety of skill-transfer scenarios, including table-tennis, kendo, Darts, and Painting. Therefore, future work should extend LinkForm by exploring more application domains, investigating mobile or wearable robot, and evaluate using video or optical see through systems based on LinkForm [6].

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